Comparative Assessment: ARABICMMLU vs. MMLU

Deployment context: Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

Deployment Context

Use case: Support foreigners visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA in answering their general questions

Target population: Non-Arab tourists or expats visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA and have general questions about the Arabic language, history or geography of these countries

Overall Risk Ratings

Benchmark	Risk	Avg. Score
ARABICMMLU	HIGH	2.0 / 5
MMLU	HIGH	1.5 / 5

Dimension Scores Comparison

Dimension	ARABIC MMLU	Rating	MMLU	Rating	Δ
Input Ontology (IO)	3/5	Moderate gaps	1/5	Serious concern	+2
Input Content (IC)	2/5	Significant gaps	1/5	Serious concern	+1
Input Form (IF)	3/5	Moderate gaps	3/5	Moderate gaps	0
Output Ontology (OO)	1/5	Serious concern	1/5	Serious concern	0
Output Content (OC)	2/5	Significant gaps	1/5	Serious concern	+1
Output Form (OF)	1/5	Serious concern	2/5	Significant gaps	-1
Average	**2.0**		**1.5**		**+0.5**

Δ = ARABICMMLU score − MMLU score. Positive = regional benchmark scores higher.

Overall Summaries

ARABICMMLU (risk: HIGH)

ArabicMMLU is a serious ground-up Arabic-language knowledge benchmark whose subject taxonomy and authentic source materials provide reasonable construct validity for evaluating school-curriculum knowledge of Arabic language, history, and geography. For the specific deployment of non-Arab tourists/expats across eight Arabic-speaking countries seeking educational-level knowledge, however, two structural mismatches dominate the assessment: (1) the single-correct-answer MCQ output ontology and form fundamentally cannot evaluate the multi-perspective, explanatory natural-language responses the deployment explicitly requires [Q27, Q88, elicitation A3]; and (2) input content and verification skew heavily Jordanian, with Morocco's school-curriculum content largely absent, Kuwait effectively absent, and no Palestinian annotators despite Palestine being a top-3 source country [Q31, Q89, WEB-3, DATASET-D4, DATASET-D5]. The 96% answer-key verification ceiling [Q43] likely reflects Jordanian/Egyptian consensus rather than pan-regional validity for contested historical and civic content. Strengths include authentic exam sourcing across eight countries, level-appropriate subject coverage, MSA text-only format matching the deployment modality, and confirmed substantial Palestinian STEM/curriculum content.

MMLU (risk: HIGH)

MMLU is fundamentally misaligned with this deployment across the four HIGH-priority dimensions (IO, IC, OO, OC). Documentation, web evidence, and dataset analysis converge on a clear finding: MMLU's 57 subjects are anchored in US academic curricula with no dedicated Arab-region content; ground-truth labels reflect US-academic and (in moral_scenarios) explicitly US-2020 framings; and the single-answer MCQ schema is structurally incapable of representing the multi-perspective output behavior required for contested Arab historical and political topics. The two MODERATE-priority dimensions (IF, OF) match deployment modality at the surface (English text) but cannot evaluate the Arabic-learning sub-cohort or open-ended explanatory output. Universal STEM and formal-reasoning content remains usable as a transferable competence baseline, but the core deployment domain — Arab history, regional geography, Arabic language basics, and Islamic religious knowledge — is essentially uncovered.

Per-Dimension Justification Comparison

Each benchmark's full assessment justification and cited evidence, shown side by side.

Input Ontology (IO)

Whether the benchmark's test case categories cover the query types expected in deployment.

ARABICMMLU — 3/5 (Moderate gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
The benchmark's taxonomy of 40 tasks across school subjects and education levels [Q25] aligns well with the deployment's school-curriculum knowledge scope (history, geography, Arabic language, civics) [Q72, Q73]. However, several taxonomic issues materially harm content validity for this deployment: (a) the taxonomy includes categories irrelevant to the tourist/expat school-knowledge use case — Driving Test, professional Law (Morocco-only), vocational General Knowledge — that were observed in the sampled data [DATASET-D5, DATASET-D26, DATASET-D27]; (b) university-level content is only 6% of the dataset [Q71]; (c) Civics content as instantiated is overwhelmingly Jordanian-specific rather than pan-Arab [DATASET-D4]; (d) Morocco's school-curriculum (history/geography/civics) is essentially absent from the sample, with Morocco's contribution concentrated in professional law [DATASET-D5]; (e) Kuwait does not appear at all in the sample [Limitations.5]. The future-work acknowledgment of needed extensions — short-answer questions, larger regional coverage, professional-domain content [Q88] — confirms these are recognized gaps.	MMLU's 57-subject taxonomy is anchored in US academic curricula with no dedicated Arab history, Middle Eastern geography, Arabic language, or Islamic jurisprudence subjects. Dataset analysis confirms that even subjects nominally relevant to the deployment (high_school_world_history, high_school_geography, global_facts, world_religions) contain little to no Arab-region content in sampled examples. Given IO is rated HIGH priority and the deployment's core domain is precisely what MMLU omits, this is a fundamental construct underrepresentation.

ARABICMMLU evidence	MMLU evidence
[Q25] 'ArabicMMLU is an Arabic multiple-choice question-answering dataset comprising 40 tasks spanning a wide range of subjects and education levels.' (p.3)	
[Q71] 'This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU' (p.7)	
[Q72] 'We present the performance of open-source models on selected subjects that potentially contain Arabic-specific contexts.' (p.7)	
[Q88] 'ArabicMMLU can be extended to include short-answer or essay questions, different modalities, larger region coverage, and more questions in professional settings.' (p.9)	

DATASET-D4: All 5 sampled Civics MS items are exclusively Jordan-specific — confirms Jordan-skew in the civics taxonomy slot	
DATASET-D5: All 5 Morocco samples are professional law, none from school-curriculum subjects	
DATASET-D26: Driving Test items (UAE, Lebanon, Egypt) — out of scope for school-curriculum deployment	
DATASET-D27: Vocational General Knowledge items (sheet metal screws) — irrelevant to tourist/expat use case	

Input Content (IC)

Whether individual datapoint content is culturally and contextually appropriate for the target region.

ARABICMMLU — 2/5 (Significant gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
Authentic ground-up sourcing from school exams across eight countries [Q1, Q26] is a substantial strength over translation-based prior work [Q5, Q6], and Palestine being a top-3 source country is materially better than the scaffold anticipated [WEB-3, WEB-4]. However, content distribution is severely unequal: Jordan contributes 6,000+ questions while some countries contribute as few as 100 [Q89]; Morocco lacks annotators and Maghrebi school-curriculum coverage [DATASET-D5]; Kuwait is effectively absent [Limitations.5]; and contested historical content is framed through Jordanian curriculum perspective [DATASET-D1, DATASET-D2, DATASET-D3]. OCR corruption in some Egyptian university items [DATASET-D23, DATASET-D24] introduces construct-irrelevant variance. Islamic Studies content is presented prescriptively from an internal Muslim perspective [DATASET-D14, DATASET-D35], misaligned with the non-Muslim tourist target population. Per the elicitation, pan-Arab framing is acceptable [A2], but the system should flag country-specific content — yet the benchmark's ground-truth treats Jordanian framing as the silent default for civic and historical questions [DATASET-D4].	All identified MMLU source materials are Anglophone Western institutions (GRE, USMLE, US undergraduate courses, Oxford University Press, Harvard Law Library) [Q23–Q25, Q83, Q85]. No Arab educational systems, regional universities, or Arabic-language curricula are represented. Dataset analysis confirms the moral_scenarios subject explicitly anchors ground truth to 'ordinary moral standards in the US as of 2020' (D12–D14) — a direct cultural-frame conflict for a deployment serving users in Arab countries. Web evidence (CAMeL, MENAValues) empirically confirms LLMs trained on such data exhibit Western/Anglocentric bias in Arab contexts.

ARABICMMLU evidence	MMLU evidence
[Q1] 'we present ArabicMMLU, the first multi-task language understanding benchmark for the Arabic language, sourced from school exams across diverse educational levels in different countries spanning North Africa, the Levant, and the Gulf regions.' (p.1)	[Q23] 'The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online.' (p.3)
[Q26] 'The questions are sourced from eight different countries in North Africa (Morocco and Egypt), the Levant (Jordan, Palestine, and Lebanon), and the Gulf (UAE, Kuwait, and KSA).' (p.3)	[Q25] 'It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books.' (p.3)
[Q89] 'ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all.' (p.9)	[Q85] 'We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law' (p.8)

[WEB-3] Jordan, Egypt, Palestine named as top-3 source countries	DATASET-D12: moral_scenarios explicitly anchors ground truth to 'ordinary moral standards in the US as of 2020'
[WEB-4] HuggingFace dataset card confirms Palestine as top-3 contributor	
DATASET-D1: Question on King Abdullah I and Zionism — Jordanian curriculum framing of contested Palestinian history	
DATASET-D4: All 5 Civics MS items are Jordan-only, confirming Jordan-skew in civics domain	
DATASET-D5: Morocco's representation concentrated in professional law, absent from school-curriculum subjects	
DATASET-D14: Islamic Studies questions presuppose Muslim reader perspective	
DATASET-D23, DATASET-D24: Severe OCR corruption in Egyptian university items	
DATASET-D35: Arabic Language reading passage frames Islamic ritual prescriptively	

Input Form (IF)

Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

ARABICMMLU — 3/5 (Moderate gaps) [high confidence]	MMLU — 3/5 (Moderate gaps) [high confidence]
The benchmark is uniformly text-only MSA [Q2, Q92], aligning with the deployment's text-only modality [A4] and MSA target [elicitation]. Multimodal content was discarded [Q37] and contextual passages preserved where needed [Q38]. However, the benchmark's MSA is fluent native-speaker register from school exams, while the deployment's users are non-native learners with basic-to-intermediate MSA proficiency, likely producing grammatically imperfect or code-switched Arabic [elicitation, expected_arabic_proficiency]. This signal-distribution gap is structural and undocumented as a design trade-off in the paper. The benchmark also notes that real-world Arabic LLM exposure includes dialectal Arabic, which is excluded here [Q92]. One minor data-quality issue observed: an Arabic Language Grammar item with English question stem [DATASET-D16] inconsistent with the stated Arabic-only design [Q30].	MMLU's text-only English MCQ format [Q3, Q56, Q71] matches the deployment's primary interaction modality (English text), which the elicitation rates MODERATE priority. However, the Arabic-learning secondary cohort cannot be evaluated by MMLU at all, and MENAValues research [WEB-11] documents cross-lingual value shifts indicating that English MMLU performance does not predict Arabic-prompt behavior. Format brittleness (e.g., separator-token sensitivity [Q105]) adds modest external-validity concern but is not region-specific.

ARABICMMLU evidence	MMLU evidence
[Q2] 'Our data comprises 40 tasks and 14,575 multiple-choice questions in Modern Standard Arabic (MSA) and is carefully constructed by collaborating with native speakers in the region.' (p.1)	[Q3] 'We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings.' (p.1)
[Q30] 'When designing ArabicMMLU, we excluded questions in English and only included questions in Arabic.' (p.3)	[Q71] 'Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format.' (p.7)

[Q37] 'workers were instructed to include only questions accompanied by an answer key, and to discard questions containing multi-modal information' (p.4)	[Q56] 'We feed GPT-3 prompts like that shown in Figure 1a... All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D,"' (p.6)
[Q92] 'The dataset primarily focuses on Modern Standard Arabic (MSA). However, multilingual and Arabic LLMs are often exposed to both MSA and dialectical Arabic.' (p.9)	[Q105] 'If we remove the </s> from the input, accuracy declines by several percentage points.' (p.12)
DATASET-D16: English question stem in Arabic-language config — minor inconsistency with stated Arabic-only design	[WEB-11] MENAValues documents Cross-Lingual Value Shifts — same model gives drastically different responses in Arabic vs. English

Output Ontology (OO)

Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

ARABICMMLU — 1/5 (Serious concern) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
This is the most critical structural mismatch for this deployment. The label schema is exclusively single-correct-answer MCQ with 2–5 candidate options [Q27], and every sampled example confirms this format [DATASET-D1, all]. The deployment explicitly requires multi-perspective acknowledgment of contested historical and civic questions [elicitation A3], but the benchmark's output taxonomy structurally cannot represent or score such pluralistic responses. The four prompt-configuration variants tested vary only in prompt language and output script [Q47], not in label structure. Contested questions like King Abdullah I and Zionism [DATASET-D1], GCC founding states [DATASET-D25], and 'elements of a state' with 'occupation' as a distractor option [DATASET-D18] are all reduced to single correct labels reflecting one curriculum's framing. The future-work note proposing short-answer and essay questions [Q88] tacitly acknowledges this limitation. Open-ended Arabic evaluation alternatives exist (AraGen, 'Beyond MCQ' — [WEB-5, WEB-6]) but are not part of ArabicMMLU itself.	MMLU's single-answer four-option MCQ schema [Q49, Q56] is structurally incompatible with the deployment's required output behavior — acknowledging multiple perspectives on contested topics (1948 war/Nakba, Western Sahara, Lebanese Civil War, Palestinian statehood, Jerusalem status), per elicitation Q3 and the contested-topics YAML section. Every sampled MMLU example has exactly one labeled correct answer with no mechanism for multi-perspective or uncertainty-flagging responses. The authors note near-random model accuracy on 'morality and law' [Q8, Q14] but do not address the structural taxonomy issue. No identified Arab-focused benchmark uses multi-perspective scoring (FV5). Given OO is HIGH priority, this is a fundamental structural-validity violation.

ARABICMMLU evidence	MMLU evidence
[Q27] 'Each question has 2–5 candidate answers, with one correct answer.' (p.3)	[Q49] 'To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.' (p.5)
[Q47] 'We initially conducted experiments with four settings... vary only in prompt language and output script' (p.5)	[Q56] 'The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction.' (p.6)
[Q88] 'For future work, ArabicMMLU can be extended to include short-answer or essay questions' (p.9)	[Q8] 'Worse, they still have near-random accuracy on some socially important subjects such as morality and law.' (p.1)
[WEB-5] 'Beyond MCQ' Arabic open-ended benchmark identified as supplementary option	[Q14] 'The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality.' (p.2)
[WEB-6] AraGen with 3C3H LLM-as-judge metric for open-ended Arabic generation	

DATASET-D1: Contested Jordanian/Palestinian history reduced to single Jordanian-curriculum correct answer	
DATASET-D18: 'Occupation' (الاحتلال) used as distractor in 'elements of a state' — politically loaded MCQ framing without multi-perspective acknowledgment	
DATASET-D25: GCC founding states reduced to single true/false answer	

Output Content (OC)

Whether ground-truth labels align with the judgments of regional stakeholders.

ARABICMMLU — 2/5 (Significant gaps) [high confidence]	MMLU — 1/5 (Serious concern) [high confidence]
Label-correctness verification process — 96% inter-annotator agreement on 100 sampled questions [Q42, Q43] with native MSA speakers holding Bachelor's degrees and competitive compensation [Q35, Q36, Q39] — establishes a reasonable practical ceiling [Q44]. However, annotator-pool composition undermines convergent validity for the multi-country deployment: 10 annotators are from Jordan (4), Egypt (2), Lebanon (1), UAE (1), KSA (2), with no Moroccan, Kuwaiti, or Palestinian annotators [Q31] despite Palestine being a top-3 source country [WEB-3, WEB-4]. For contested historical questions where 'correct' answers vary across national curricula, the 96% agreement likely reflects Jordanian/Egyptian consensus rather than pan-regional validity. The sampled data shows concrete examples of Jordanian-curriculum framing for shared Arab history [DATASET-D1, DATASET-D2, DATASET-D3, DATASET-D11]. Training data contamination is acknowledged as unverifiable [Q95]. OCR corruption in university Egyptian items [DATASET-D23, DATASET-D24] suggests label quality may degrade in some sub-corpora. No aggregation method considerations for minority perspectives are documented.	MMLU annotator population is graduate/undergraduate students from US institutions [Q23] with Mechanical Turk and US 95th-percentile test-taker baselines [Q31, Q33, Q34] — no Arab regional stakeholders. Ground-truth labels reflect Western academic framing, with moral_scenarios explicitly encoding 'US standards as of 2020' (D12). Inter-annotator agreement is not documented. Empirical evidence (CAMeL, ILMAAM) confirms Western framing of Arab content in LLM evaluation systematically diverges from Arab stakeholder judgments [WEB-5, WEB-15]. With OC rated HIGH and the deployment serving eight Arab countries with contested historical narratives, this is a major convergent- and external-validity violation.

ARABICMMLU evidence	MMLU evidence
[Q31] 'a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian).' (p.4)	[Q23] 'The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online.' (p.3)
[Q42] 'we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions' (p.4)	[Q31] 'Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test.' (p.3)
[Q43] 'We found that 96% of the questions and answer keys match on average' (p.4)	[Q33] 'For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations' (p.3)
[Q44] 'This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU.' (p.4)	[Q34] 'If we take the 95th percentile human test-taker accuracy for exams that build up our test... we then estimate that expert-level accuracy is approximately 89.8%.' (p.3)

[Q95] 'we cannot assert that the model's pretraining data is free from contamination.' (p.10)	[WEB-5] ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity
[WEB-3] Palestine confirmed top-3 source country — yet no Palestinian annotators in pool	[WEB-15] Naous et al. 2024 (CAMeL) empirically demonstrate LLM Western cultural bias in Arabic prompts
[WEB-4] HuggingFace dataset card confirms Palestine major contributor	DATASET-D12: moral_scenarios labels explicitly encode 'ordinary moral standards in the US as of 2020' as the ground-truth criterion
DATASET-D1: Jordanian-framing of King Abdullah I/Zionism question — verified by non-Palestinian workers	
DATASET-D3: Jordanian-British 1928 treaty framing as authoritative	
DATASET-D11: Jordanian royal history as primary social science fact	
DATASET-D23, DATASET-D24: OCR-corrupted Egyptian university items where label correctness becomes ambiguous	

Output Form (OF)

Whether the expected output modality matches regional deployment needs and accessibility.

ARABICMMLU — 1/5 (Serious concern) [high confidence]	MMLU — 2/5 (Significant gaps) [high confidence]
The benchmark measures performance as percentage MCQ accuracy [Q3, Q45] via highest-token-probability scoring for open-source models [Q50, Q79] or first-token regex matching for closed-source models [Q53, Q54]. This output form is fundamentally misaligned with the deployment's requirement for open-ended, explanatory, multi-perspective natural-language responses [elicitation A3]. The benchmark cannot evaluate whether a model produces appropriately nuanced, perspective-flagging answers — it can only score whether the model selects the curriculum-designated correct option. Additional output-form analyses (calibration [Q80], length-confidence correlation [Q82], few-shot trends [Q75, Q76], negation sensitivity [Q83, Q84]) all operate on the same MCQ scoring surface. The text-based MCQ form is appropriate for literacy-capable populations [Q93] in principle, but the form mismatch with the deployment's natural-language explanatory output requirement is structural and not addressable through reweighting or filtering. Supplementary frameworks (AraGen 3C3H, 'Beyond MCQ') exist [WEB-5, WEB-6] but are separate evaluations.	Both MMLU and the deployment use text output, satisfying basic modality match. However, MMLU's classification-accuracy metric over A/B/C/D tokens [Q49, Q56] cannot evaluate the open-ended, multi-perspective explanatory responses the deployment's tourist/expat assistant produces. Calibration evaluation [Q64–Q67] is a useful auxiliary signal — GPT-3's 24% confidence-accuracy gap [Q66] is informative — but does not address the form mismatch. With OF rated MODERATE, this is a real but partial concern: MMLU can score factual recall in MCQ form but not the explanatory output form the deployment expects.

ARABICMMLU evidence	MMLU evidence
[Q3] 'Our comprehensive evaluations of 35 models reveal substantial room for improvement' (p.1)	[Q49] 'To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.' (p.5)
[Q45] 'Our experiments focus on zero-shot and few-shot settings across 35 models.' (p.5)	[Q56] 'The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction.' (p.6)

[Q50] 'for open-source models, we determine the answer based on the highest probability among all possible options.' (p.6)	[Q66] 'Its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.' (p.7)
[Q53] 'For closed-source models, we determine the answer based on the first token generated in the text using a regular expression.' (p.6)	
[Q93] 'ArabicMMLU is focused solely on text-based assessment, and the exploration of multimodal questions is left for future work.' (p.9)	
[WEB-5] 'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF	
[WEB-6] AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation	